

Scientific Review & Synthesis

Document Type: Foundational Study | **Year:** 2012

DOI Reference: [10.1016/j.scitotenv.2012.03.084](https://doi.org/10.1016/j.scitotenv.2012.03.084)

Plant species differences in particulate matter accumulation on leaf surfaces

Authors: Arne Sæbø, Robert Popek, Bianka Nawrot, Hans Martin Hanslin, Helena Gawro G6¶ Â Stanis & w W. Gawro G6¶•

1. Abstract & Scope

Poor urban air quality continues to pose severe risks to public health. Deploying vegetation serves as a highly effective complementary remediation strategy. Tree canopies naturally promote air turbulence, which, combined with specific leaf surface traits, facilitates the deposition and retention of particles. This study evaluated a cohort of 47 woody species (22 trees and 25 shrubs) grown at high-exposure motorway and low-exposure rural plots in Norway and Poland. Epicuticular wax was isolated to separate Surface PM (SPM) and Wax-embedded PM (WPM). Results document that accumulation varies 10- to 20-fold among common species. Total epicuticular wax volume was the strongest predictor of ultrafine (PM $\leq 0.1 \mu\text{m}$) particle trapping, while foliar trichomes (hairs) enhanced coarse (PM $\geq 2.5 \mu\text{m}$) capture but had no significant effect on sub-micron particles.

2. Methodology & Experimental Parameters

Foliage agitated in water (SPM) followed by chloroform rinse (WPM) to solubilize epicuticular waxes. Cascade filtration on Type 91 (10 μm), Type 42 (2.5 μm), and PTFE (0.2 μm) membranes. Normalization via digital scanning.

3. Core Findings & Data Synthesis

High-capacity taxa (*Betula pendula*, *Pinus mugo*, *Pinus sylvestris*) accumulated 24-55 $\mu\text{g}/\text{cm}^2$ total PM, while low-capacity taxa (*Acer platanoides*, *Tilia cordata*) were limited to 6-13 $\mu\text{g}/\text{cm}^2$. Epicuticular wax volume dominated PM $\leq 0.1 \mu\text{m}$ capture.

4. Strategic Relevance to JU and Kolkata Planning

Provides the benchmark parameters for waxy evergreens vs flat-leaved deciduous trees, highlighting *Betula pendula* and *Pinus* species as elite monitoring templates.

